

Module 11

Costs for Decision Making

Learning Objectives

By the end of this module you should understand:

1. The distinction between full and variable costing
2. The impact on reported profit between the two cost systems
3. The concept of relevant cost for special decisions
4. The analytical framework surrounding relevant costs
5. The application of relevant costs to:
 - the make or buy decision
 - opportunity cost
 - sunk cost
 - the special sales order
 - further processing

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11.3 Managerial Implications of Absorption versus Variable Costing

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Case Studies

In module 9, we've covered the behavior of different types of cost; where costs are broken down into variable and fixed costs

Variable Costs

Costs that change in magnitude with production or sales levels

Fixed Costs

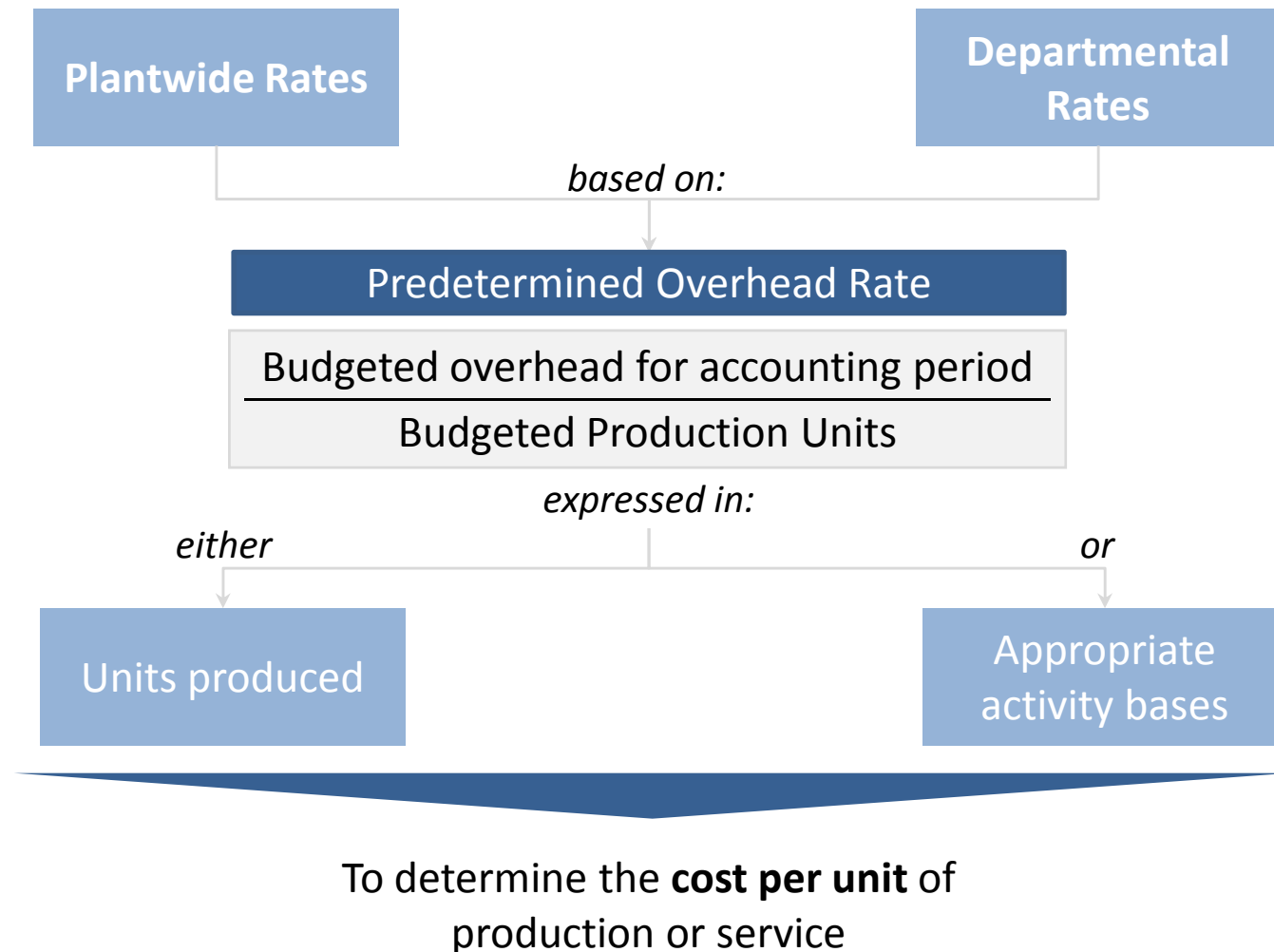
Costs that do **NOT** change as production or sales move up or down

provide:

Useful Managerial
Information

e.g. break-even points & the impact on profit of changing sales mixes

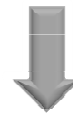
Module 10 described various procedures for spreading those costs using plantwide and departmental overhead rates



This calculation is required before actual production and sales take place so that management can make decisions on output levels and prices

This module explores the implications of combining the desirable features of both of these topics

- 1 Calculating the full cost per unit
- 2 Managerial decisions which lie outside normal routine procedures



So that:

Fundamental
management choices
can be made on
appropriate costing
system

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Case Studies

In the DVD cabinet example, we've allocated the share of manufacturing overhead and that of non-manufacturing overhead through the predetermined rate

One timber-shelved DVD unit

	€
Direct material:	
Chipboard	10.00
Veneer	3.00
Metal trim	2.00
	15.00
Direct labour 1.5 hours @ €5 per hour	7.50
Prime cost	22.50
Manufacturing overhead	2.50
Manufacturing cost	25.00
Share of non-manufacturing overhead	5.00
Total cost	30.00

**Predetermined
Overhead Allocation
Rate**

=

Budgeted overhead for accounting period

Budgeted Production Units

Such a calculation must be made in advance of production so that the desired selling price can be calculated

The determination of both numerator and denominator is fraught with problems, but that of the denominator is usually more unpredictable

Numerator

Sometimes overheads fluctuate as their market prices may vary during the accounting period

But if:

The estimation process has been handled carefully

then

The variations at the end of the period are usually small

Denominator

The denominator is not so readily predictable

Business may plan to produce and sell 1,000 units per year based on experience, a figure determined by market research exercises and production capacity

but

A sudden upsurge in demand can cause the actual number produced and sold to go to 1,200 units

and/or

An unforeseen shortage in raw materials could reduce production to 800 units

The denominator would cause the predetermined allocation rate to be either too low or too high

1) Example showing that when actual production equals planned production, no denominator volume variances will occur

Example

A business manufacturing designer fountain pens planned to produce and sell 10,000 in 20x2. Costs of manufacture per pen were:

	€
Direct material	
Nib, casing, reservoir mechanism	20
Direct labour	
Craftsmen, machine minders	30
Variable costs of production	<u>50</u>
Fixed production costs	
Depreciation, management salaries, R&D, maintenance, etc.	
Budgeted total €100 000/10 000	<u>10</u>
Full production cost	<u><u>60</u></u>

Here the allocation of fixed production overheads was based on the budgeted total expenditure and the budgeted volume of output of pens for 20x2, namely 10,000. Each pen sells for €100 yielding a gross profit of €40 per pen from which selling, administrative and financial costs would be deducted. Assume these costs to amount to €200,000.

Example (Continued)

If in 20x2 everything went according to plan the business would produce a P/L account like this:

	€
Sales 10 000 pens @ €100	1 000 000
Less: Full cost of sales 10 000 @ €60	600 000
Gross profit	400 000
Other expenses	200 000
Net profit	<u>200 000</u>

This form of accounting is called full or absorption cost accounting because each unit is asked to absorb its share of fixed production overhead, a reasonable requirement

Example (Continued)

However, let us assume that management's freedom of pricing was severely restricted because of competitive forces in the designer fountain pen market; in addition because of uncertainty of demand the business produced only what was required and therefore made no attempt to allocate fixed production overhead to each product.

If, during 20x2, the company achieved exactly the same outcome, that is, producing and selling 10,000 pens with the same costs the profit and loss account would look like this:

	€	€
Sales 10 000 pens @ €100		1 000 000
Less: Variable cost of sales 10 000 @ €50		500 000
Contribution margin		500 000
Fixed production costs	100 000	
Other expenses	200 000	300 000
Net profit		<u>200 000</u>

This form of accounting is called variable or direct costing because it accounts for fixed costs in a lump sum

Note that contribution is a term used to show the sales revenue less variable (marginal) costs incurred in generating this revenue

Contribution Margin

$$\begin{array}{r} \text{Sales Revenue} \\ - \text{Variable (marginal) costs incurred} \\ \hline \text{Contribution Margin} \end{array}$$

We conclude that the net profit under both methods is identical; that's because actual production equals planned production and sales equals production

If:

Actual production equals planned production

Sales equals production



Net profit under both methods is identical

But we know that actual production hardly ever matches exactly planned production, and it is rare to find a business where all production is sold by the end of the accounting period

2) Example that shows that bottom-line profit difference arises when sales is not equal to actual production

Example

Assume that the absorption costing method is applied to the fountain pen business described above. Planned production was 10,000 but actual production was 11,000 due to a desire to keep operatives busy throughout the year. The excess will be sold next year. Throughout the year each pen will be allocated €10 fixed production costs as before but in the light of actual production level achieved this should have been:

$$\frac{€100\,000}{11\,000} = €9.09$$

Each unit has over-absorbed €0.91, a total of €10,000 (11 000 × €0.91) for the year

Absorption method produces higher net profit in P/L account, though it's been adjusted for the denominator volume variance at the end of the year

This over-absorbed sum (€10,000) has been caused by the wrong denominator volume being selected and must be corrected at the end of the year, otherwise the production would be seen to be absorbing more costs than had been incurred.

Absorption profit and loss account	€	€
Sales 10 000 pens @ €100		1 000 000
Less: Full cost of sales 10 000 @ €60		<u>600 000</u>
Gross profit		400 000
Denominator volume variance	(10 000)	
Other expenses	<u>200 000</u>	<u>190 000</u>
		<u>210 000</u>

Note that the denominator volume variance helps to reduce production costs in the P/L account which otherwise would be overstated by the over-absorption of fixed production costs

We set out the two methods (absorption and variable costing) side by side

The variable (or direct) costing profit and loss would be identical to the one already produced. We set out the two methods side by side

Absorption costing			Variable costing		
	€	€		€	€
Sales		1 000 000	Sales		1 000 000
Less: Full cost of sales		600 000	Less: Variable cost of sales		500 000
Gross profit		400 000	Contribution margin		500 000
Denominator volume variance	(10 000)		Fixed production expenses	100 000	
Other expenses	200 000	190 000	Other expenses	200 000	300 000
Net profit		210 000	Net profit		200 000

Why the difference of €10,000?

1,000 pens have been produced which have not been sold; these pens have been placed in inventory, each one valued, under absorption costing, at €60, i.e. €60,000. Under variable costing the same 1,000 pens are valued at €50 each, i.e. €50,000. Therefore under absorption costing an extra €10,000 of cost has been held back in inventory for release in the next period

Note that the difference in bottom-line profit has nothing to do with the denominator volume variance

On the other hand the absorption method produces a different figure (less) when the actual production is less than the planned one

We move our example one year on and keep all facts and figures the same except that production is reduced to 9,000 pens even though the estimate at the beginning of the year was again 10,000. For 20x3 the two profit and loss accounts would be:

Absorption costing			Variable costing		
	€	€		€	€
Sales		1 000 000	Sales		1 000 000
Less: Full cost of sales		<u>600 000</u>	Less: Variable cost of sales		<u>500 000</u>
Gross profit		400 000	Contribution margin		500 000
Denominator volume variance	10 000		Fixed production expenses	100 000	
Other expenses	<u>200 000</u>	210 000	Other expenses	<u>200 000</u>	300 000
Net profit		<u>190 000</u>	Net profit		<u>200 000</u>

Again there is a difference of €10,000 in bottom-line profit and again a denominator volume variance of €10,000. And again the two figures are unrelated

This difference in profit is due to the value of inventory needed to satisfy 20x3's sales

Denominator Volume Variance

↓ *Caused by:*

The fact that the 9,000 pens manufactured only absorbed €90,000 fixed production expenses while €100,000 was incurred in 20x3

The shortfall must be written off at the end of the year

Difference in Profits

↓ *Caused by:*

Production was 9,000; sales were 10,000; therefore 1,000 pens were taken from inventory. Under absorption costing these units emerge from inventory at €60 instead of €50 under variable costing

Profits therefore fall by $1000 \times €10$ (the difference in inventory value per pen which itself is the fixed production costs per pen)

Conclusion

A denominator volume variance



When actual production $\neq$ planned production

Bottom-line profit difference



When sales $\neq$ actual production

Work-it-yourself example

Complete the following table which works a number of permutations on the above data

As a reminder, the data are as follows.

Cost of fountain pen	€
Direct material	20
Direct labour	30
Fixed production costs*	10
Full cost	<u>60</u>

* Based on budgeted total fixed production costs of €100 000 and a budgeted level of production of 10 000 pens per annum.

Work-it-yourself example (Continued)

The table:

Absorption and variable costing compared									
Column	1	2	3	4	5	6	7	8	9
Sales (000s)	10	8	11	9	7	10	12	9	13
Production (000s)	10	8	11	10	8	11	10	8	11
Sales revenue	1 000					1 000		900	
<i>Absorption costing</i>									
Cost of sales	600	–	–	–	–	600	–	540	–
Gross profit	400					400		360	
Volume variance	–	–	–	–	–	10	–	(20)	–
Net profit	400					410		340	
<i>Variable costing</i>									
Cost of sales	500	–	–	–	–	500	–	450	–
Contribution margin	500					500		450	
Fixed costs	100	–	–	–	–	100	–	100	–
Net profit	400					400		350	
	Production = Sales			Production > Sales			Production < Sales		

Answer the following two questions before proceeding

1. In column 5, why is there a difference in profits of €10,000?
2. In column 9, why is there a difference in profits of €20,000 when the denominator volume variance is only €10,000?

Work-it-yourself example (Solution)

Solution to table on absorption and variable costing

Column	1	2	3	4	5	6	7	8	9
Sales (000s)	10	8	11	9	7	10	12	9	13
Production (000s)	10	8	11	10	8	11	10	8	11
Sales revenue	1 000	800	1 100	900	700	1 000	1 200	900	1 300
<i>Absorption costing</i>									
Cost of sales	600	480	660	540	420	600	720	540	780
Gross profit	400	320	440	360	280	400	480	360	520
Volume variance	–	(20)	10	–	(20)	10	–	(20)	10
Net profit	400	300	450	360	260	410	480	340	530
<i>Variable costing</i>									
Cost of sales	500	400	550	450	350	500	600	450	650
Contribution margin	500	400	550	450	350	500	600	450	650
Fixed costs	100	100	100	100	100	100	100	100	100
Net profit	400	300	450	350	250	400	500	350	550
	Production = Sales			Production > Sales			Production < Sales		

Work-it-yourself example (Solution)

Solution to table on absorption and variable costing

Column	1	2	3	4	5	6	7	8	9
Sales (000s)	10	8	11	9	7	10	12	9	13
Production (000s)	10	8	11	10	8	11	10	8	11
Sales revenue	1 000	800	1 100	900	700	1 000	1 200	900	1 300
<i>Absorption costing</i>									
Cost of sales	600	480	660	540	420	600	720	540	780
Gross profit	400	320	440	360	280	400	480	360	520
Volume variance	–	(20)	10	–	(20)	10	–	(20)	10
Net profit	400	300	450	360	260	410	480	340	530
<i>Variable costing</i>									
Cost of sales	500	400	550	450	350	500	600	450	650
Contribution margin	500	400	550	450	350	500	600	450	650
Fixed costs	100	100	100	100	100	100	100	100	100
Net profit	400	300	450	350	250	400	500	350	550
	Production = Sales			Production > Sales			Production < Sales		

The only scenario in which profits are the same under the two cost systems

Absorption produces higher profits, because more fixed production expenses are 'parked' in inventory. But this action must be reversed because of cash flow problems (cash required for production not being restored by sales)

Vice versa

Work-it-yourself example (Solution)

1

In column 5 the business has sold 7,000 pens but produced 8,000. The 1,000 pens placed in inventory are valued at €60,000 under absorption costing and €50,000 under variable costing. Therefore the variable costing profit and loss account must bear €10,000 more cost.

2

The two figures are unconnected. The favorable denominator volume variance arises because the business has produced 11,000 pens instead of 10,000 as planned and has therefore over-absorbed €10,000 of fixed production expenses. The difference in the two bottom-line profits is due to the fact that 2,000 pens had to be taken from inventory to satisfy the buoyant sales of 13,000. Each pen taken from inventory was €10 more expensive under absorption costing than under variable costing

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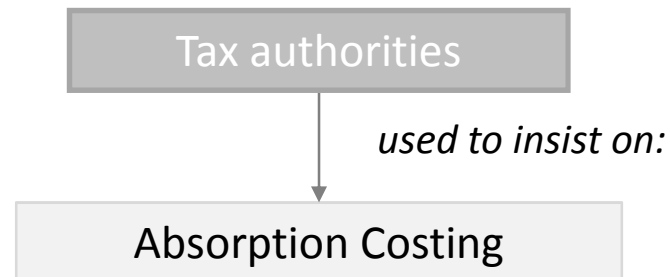
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It is sometimes argued that the choice of using absorption or variable costing is not open to a business as many tax authorities insist on absorption costing



Inventory valuations are consistently higher and therefore taxable profits are higher

But with the advent of highly flexible and powerful information systems the requirements of special user groups (e.g. the tax authorities) can be catered for separately from managerial requirements

The sole criterion for choice must be: which method flags up the clearer set of signals?

	Variable Costing	Absorption Costing
Used By:	Managers who appreciate the distinction between variable and fixed costs and the role of contribution margin	Other managers
Price Reduction	Sales people, if supplied with variable cost information, may be tempted to reduce prices to secure more orders	It would prevent price reductions in a situation of competitive stress where variable cost sales are better than no sales at all
Profit Influence	Profit is influenced not only by sales but also by production levels	Profit is only a function of sales, so when sales rise, so do profits
Fixed Expenses	Tucks away some into inventory but then eventually release them in a succeeding accounting period	Recognizes them all every accounting period

The difference between absorption and variable costing is simply a question of timing of fixed production expenses

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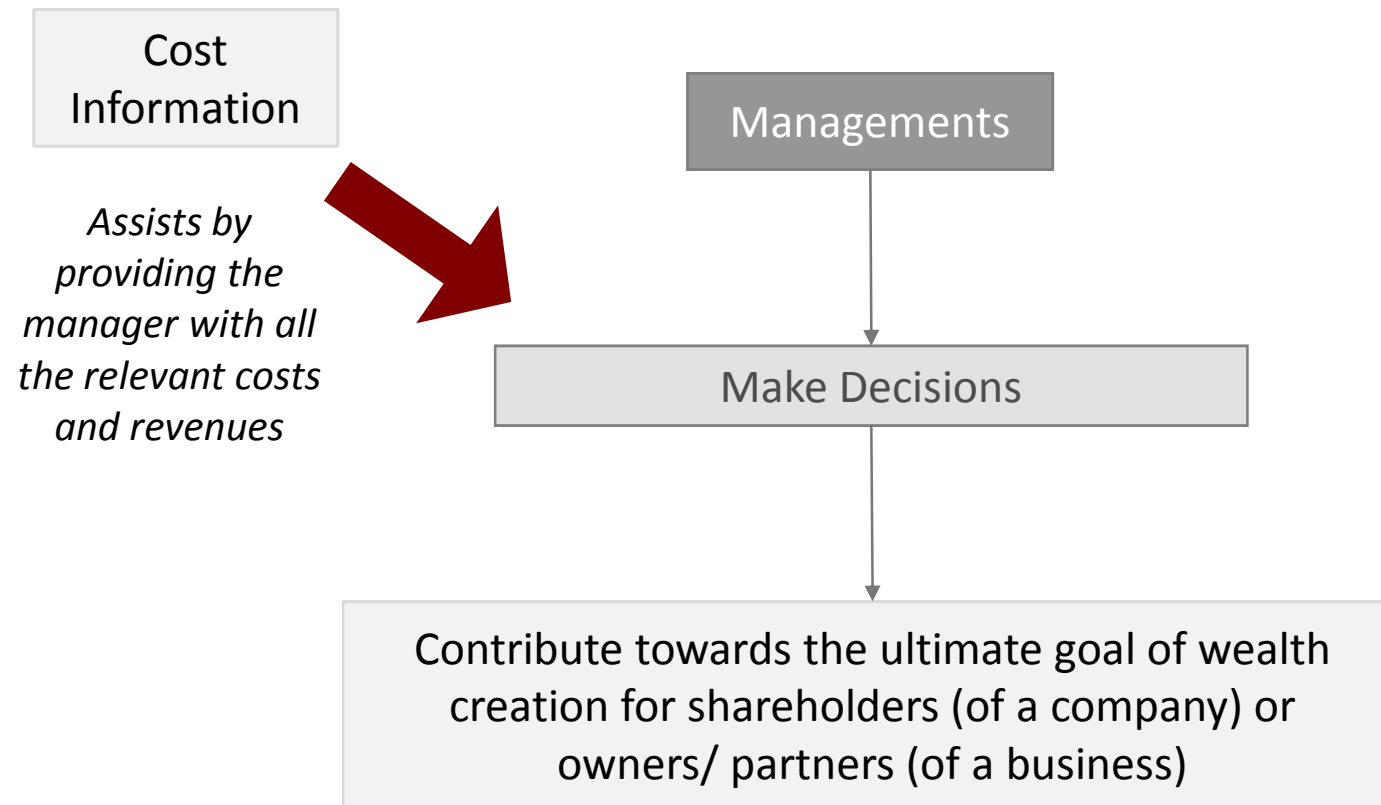
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In the section to come, we will address which costs and revenues are relevant & which are not relevant to the manager in his task of making decisions?



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In general terms, two major categories of decision can be identified, routine decisions and special, or non-routine, decisions

Routine Decisions

Decisions a manager takes in the normal course of events; depends on the manager's personality & facility in dealing with people



- Sorting out disruptions to production caused by machine failure
- Labor problems or breakdown of supplies
- Dealing with awkward customers



Cost information (if required) will be extracted from the various routine cost systems described earlier

Non-Routine Decisions

Decisions that require less personal, direct negotiation; require logical analysis of the facts and alternatives in a cool and rational manner, with the manager taking the decision after much thought



- Deciding whether or not to shut down a department or a product
- Whether to continue making a component or to purchase the required volume from an outside supplier



Cost data tend to play a significant part in the information which the manager needs

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Non-routine decisions are approached in a routine manner

The following list of tasks is often used by management:

- 1 Define the Problem and List All Feasible Alternatives
- 2 Cost the Alternatives
- 3 Assess the Qualitative Factors
- 4 Make the Decision

Non-routine decisions are approached in a routine manner

The following list of tasks is often used by management:

- 1 Define the Problem and List All Feasible Alternatives
- 2 Cost the Alternatives
- 3 Assess the Qualitative Factors
- 4 Make the Decision

Task one includes fully understanding what the real problem is and setting out all possible alternatives which should be considered

Let's take the following example:

Last year the management of a wholesale butchery business was confronted with a continuing trend of losses being incurred on every ton of shin beef and leg beef sold. Losses were running at such a level that the accountants suggested that these cuts of beef should be deep frozen until such time as the market demand for cheaper portions picked up.

Management accepted their advice. The problem facing management now is the shortage of deep frozen facilities and the proposal to build a new cold storage facility to stock these cuts, which are still not selling well.

On the face of it this scenario describes a fairly typical investment decision

Investment Decision

Should the company invest a substantial sum in a new cold storage plant and if so, how should the company seek to finance the project?

Other Suggestions

1

Wouldn't it be less costly to destroy surplus cheap cuts on a First In First Out basis than to build the new factory?

2

Shouldn't the company engage in a massive advertising campaign to persuade the consumer of the advantages of these cheaper cuts (good for meat loaves and for producing stock with high nutritional value) and in so doing raise the price of the cuts in an effort to cover losses?

The problem and the alternatives seem well stated and clear-cut. But the real problem lies in the costings of the various cuts of beef

Management discovers that the abattoir labor and overhead costs have been allocated to the different cuts on the basis of weight before preparation for sale without any consideration of the market value of these parts

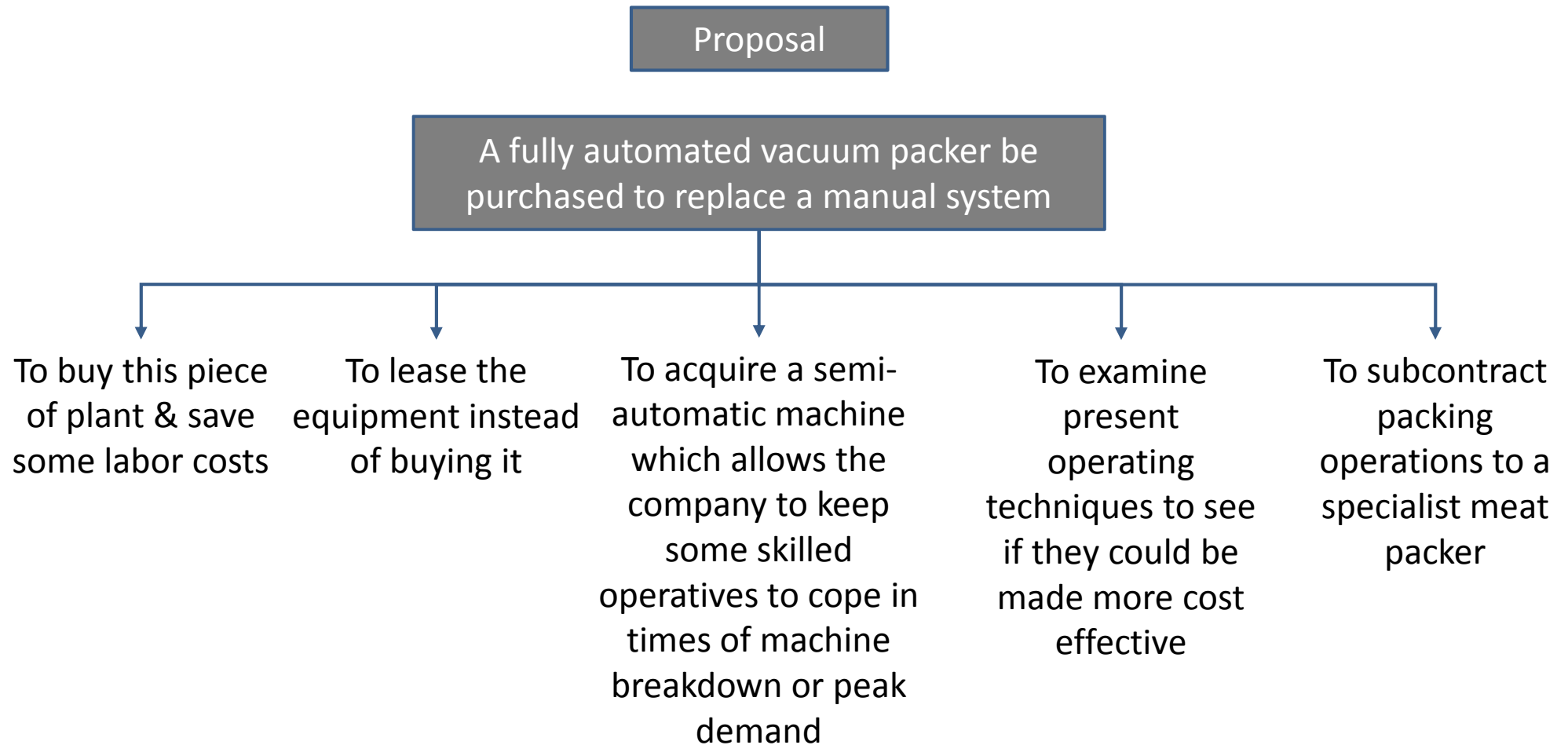


The problem identified

To devise a more appropriate allocation key for costs so that the various cuts of meat bear the overheads in a way that would not affect the decision to sell

This management problem, once identified, is much less complex or costly than the prospect of having to build new plant to store overpriced inventory

Second, management must identify all feasible alternatives



All of these alternatives would need to be examined closely before management made a decision which could have an effect on the profitability of the company

Non-routine decisions are approached in a routine manner

The following list of tasks is often used by management:

- 1 Define the Problem and List All Feasible Alternatives
- 2 Cost the Alternatives
- 3 Assess the Qualitative Factors
- 4 Make the Decision

Each alternative identified above would need to be measured in terms of cost, that is, cost outlays and cost savings

Consider the alternative to acquire a semi-automatic machine and keep some labor against the alternative to purchase a fully automated machine

The relevant costs may be expressed as follows:

	Fully automatic	Semi-automatic
Purchase price	€10 000	€6 000
Labour costs	1 000	5 500
Power costs	500	300
	€11 500	€11 800

Difference in favor of fully automatic machine is €300

The relevant costs are those that differ for the alternatives under review; administrative, rent, heat and light costs are not relevant to the packing process & the decision-making

It would be more helpful to management to list the relevant costs in terms of the differences in amount:

	Fully automatic versus semi-automatic
Increase in purchase price for fully automated plant	€(4000)
Saving in labour cost	4500
Increase in power cost	(200)
Net savings in fully automated plant	<u>€300</u>

The appropriate way of analyzing these relevant costs will be dealt with in Module 15

Non-routine decisions are approached in a routine manner

The following list of tasks is often used by management:

- 1 Define the Problem and List All Feasible Alternatives
- 2 Cost the Alternatives
- 3 Assess the Qualitative Factors
- 4 Make the Decision

Certain features that can't be measured in terms of cost may influence the decision, especially if there's a very small cost differential – these are called qualitative factors

One major qualitative factor

Relationship management enjoys with the workforce

If the business is a relatively small family concern with a small, loyal workforce, the temptation to select the semi-automatic machine which permits the retention of some workers may be overwhelming

Non-routine decisions are approached in a routine manner

The following list of tasks is often used by management:

- 1 Define the Problem and List All Feasible Alternatives
- 2 Cost the Alternatives
- 3 Assess the Qualitative Factors
- 4 Make the Decision

The function of management is to make things happen in accordance with its plans

A decision can include:

To do nothing –a perfectly valid decision

To put off the decision as a whole –because of the information required

Management must balance the costs of producing this additional information with the benefits which may be derived from the information obtained

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Task Two: Cost the Alternatives, receives more attention in this part of the course; where we will explain some basic issues to separate relevant costs from non-relevant

Costing the alternatives

From both the revenue and cost angles



Require:

Management to be concerned only with the **Relevant Costs**, not the total costs, of each alternative

Costs that change as a result of the decision being contemplated



They are called **Differential Costs**

The problem facing the accountant is how to recognize when a cost is a relevant cost

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Full costing, as described in Module 10, is an accounting system which is concerned with allocating all costs, direct and indirect, to the units of output which are to be sold

Full Costing

- Allocating all costs, direct and indirect, to the units of output which are to be sold.
- It includes direct costs (raw material, direct labor) and some of the indirect costs of both the manufacturing and administrative sectors of the business.
- Some of the indirect costs are fixed costs

A problem arises:



Fixed costs, once allocated assume the appearance and characteristics of variable costs



The more volume that is produced, the higher the costs

When total variable costs and total fixed costs are split into individual products; variable costs seem fixed while the fixed costs vary

We repeat the illustration used in Module 9:

Possible volume levels	100	200	300
Variable costs in total	€100	€200	€300
Fixed costs in total	€600	€600	€600
Variable cost per unit	€1	€1	€1
Fixed cost per unit	€6	€3	€2



When total variable costs and total fixed costs are split into individual products; variable costs seem fixed while the fixed costs vary

When choosing between alternative courses of action, management should ignore costs which will not change; fixed costs will remain fixed in total regardless of alternatives and can therefore be ignored

Although the emphasis in examining relevant costs is on variable, or marginal costs, this should not prevent a consideration of fixed costs too

Example

- One alternative could involve the acquisition, on lease, of a computer which would require a specially constructed air-conditioned facility.
- The lease payment, depreciation and rental charges associated with such a decision alternative are all relevant even although they are fixed costs.

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By taking a decision to implement a specific option, management automatically forgoes the opportunity of embarking on alternative options

Back to the Butchery example:

It may be recognized that the labor required to be retained if the semi-automatic machine is acquired would be under-used for at least three months each year. It is also thought that their services could be subcontracted to competitors during this time for €80 per month.

	Fully automated	Semi-automatic
Total relevant costs	€11 500	€11 800
Opportunity cost of forgoing purchase of semi-automatic	240	
	€11 740	€11 800

→ *If taken into account as a relevant cost it would serve to close the gap even further between the two alternatives by €60 in favor of fully automated machine*

Alternatively the opportunity 'revenue' of €240 could be deducted from the relevant costs of €11 800 for the semi-automatic machine – the result would be the same

Accountants are uncomfortable with opportunity costs because they are unclear and highly subjective

- | | | |
|---|------------|---|
| 1 | Unclear | These costs cannot be determined from the accounting system |
| 2 | Subjective | They are highly subjective and uncertain |

Managers should make every effort to quantify such costs because they could have a profound impact on the other relevant costs and therefore on the decision to be taken

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In deciding on the costs which are relevant to the decision under review, it is essential that a company-wide, not purely a departmental, stance is adopted

Back to the Butchery example:

Consider that the surplus labor capacity could be allocated to other departments during the three slack months instead of to an outside firm. Is there still an opportunity cost associated with such a shift in resources?

The answer is no, unless this saves the other departments having to employ additional people from outside. Although the meat-packing department will be able to reduce its annual wages bill (by having some other department charged for the labor 'subcontracted'), the company's wages bill is not affected one way or the other. This internal charge is not relevant. Such a charge or cost only becomes relevant if the company's overall financial position is affected.

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Relevant cost analysis must not be confused by mixing costs which have been incurred from old decisions with costs which will be incurred in the future

Sunk Costs

Costs which have already been incurred and that should be ignored when looking for relevant costs

Back to the Butchery example:

About the decision confronting the management of the wholesale butchery business:

- One director reminds his colleagues of the expensive wrapping machine that was purchased the previous year and which would have to be scrapped if a new vacuum packer was purchased. (There would now be no market for such a technically obsolete machine in the butchery trade.) His argument is that the annual depreciation charge (€1,000 per year, based on a five-year straight-line depreciation charge on a €5,000 machine) would be saved if a new vacuum packer is purchased and is therefore a relevant cost saving.
- Another director, however, expresses surprise at this conclusion: his contention is that because only €1,000 has been written off the obsolete machine, the relevant cost is €4,000, being its book value. Which of the directors is correct?

Relevant costs are future costs; relevant costs involve cash spent or cash saved

Answer

Neither of them is correct!! The wrapping machine is a sunk cost. However a large sum of money expended, and however regrettable and inappropriate the investment has turned out to be, nothing can alter the decision taken last year. Regardless of whether the fully automated machine, the semi-automated machine or neither is acquired and the one-year-old wrapping machine remains operational, the €4,000 book value will be written off one way or another. If the machine is scrapped the write-off will be immediate; if it is not scrapped the write-off will be over four years. In either case the total cost will be the same and is therefore not relevant to the decision confronting the directors.

The only relevant cost (income or negative cost) would be any sales proceeds from the disposing of the wrapping machine

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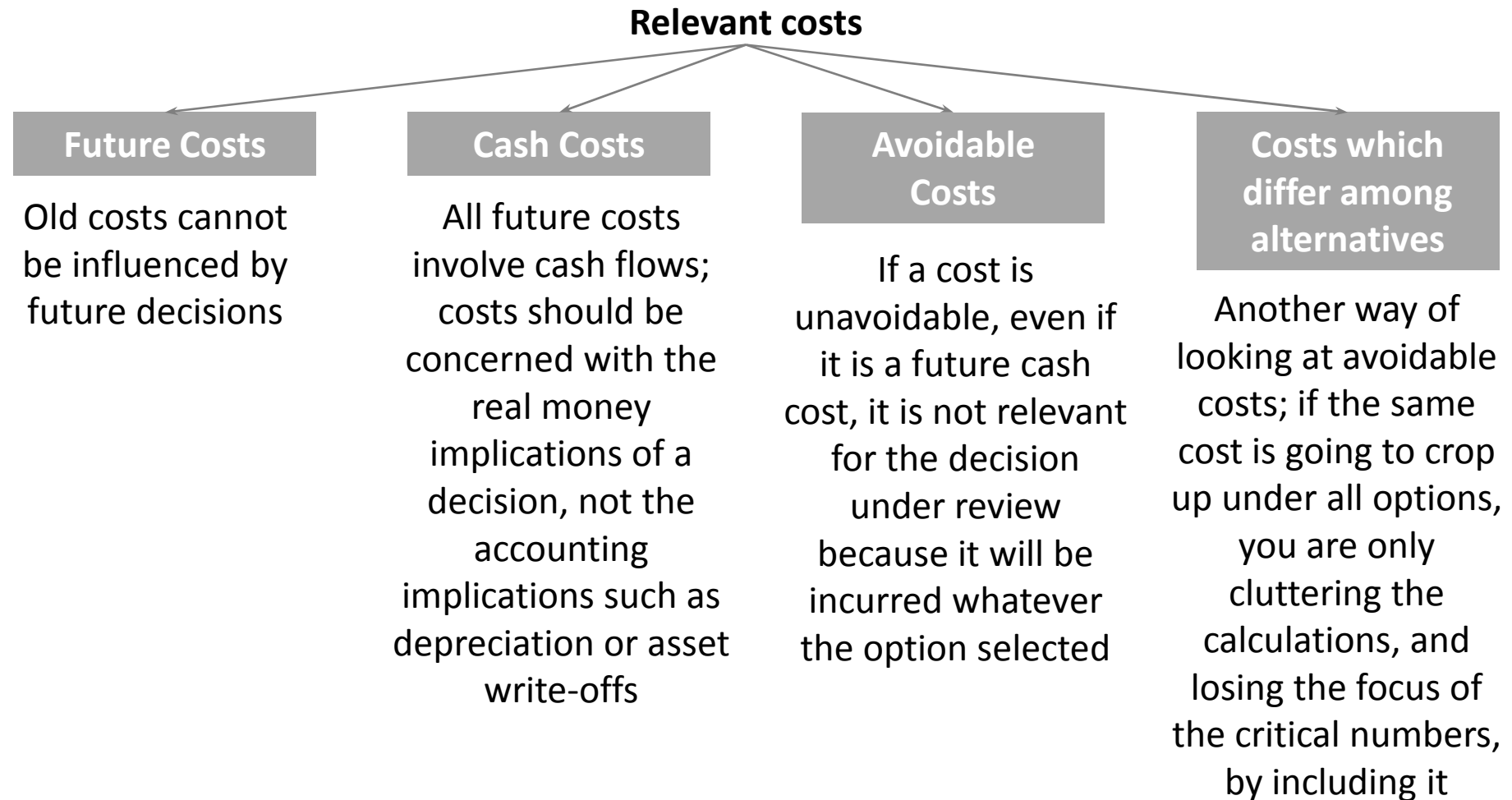
So far we've covered the replacement of labor by machine & the associated cost implications, we now select three further scenarios for analysis

Closing Down a Unit

The Special Sales Order

Should we process further?

But at first we have to set out the analytical framework so far developed



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Management is frequently confronted with the alternative of shutting down, either temporarily or permanently, a product line, or department, or factory, or subsidiary

To identify the relevant costs of such an alternative

Best way is to ask:

How long should operations be kept going?

If products make a contribution towards fixed costs then production should continue

Fixed costs may be reduced if operations were shut down, for example administrative salaries could be avoided, while some special shut-down costs may be incurred like plant maintenance, security guards' fees, redundancy and retraining costs. In such a situation, the term 'fixed costs' should be interpreted carefully

Example

The Clydeside Anchor Co. specializes in forging anchors and manufacturing ancillary equipment for oil-rigs. A world glut of oil had caused a slowing-down of oil-rig building which has had an immediate effect on all product lines of Clydeside. Particularly badly hit has been its chain division. The directors are considering whether this division, consisting of one factory, should be closed until the glut has eased.

A flexible budget has been compiled.

	Production capacity (Fixed costs + Variable costs)				Fixed costs	
	40%	60%	80%	100%	Closedown	Normal
Direct material	240 000	360 000	480 000	640 000	–	–
Direct labour	200 000	300 000	400 000	500 000	–	–
Factory overheads	200 000	220 000	240 000	260 000	120 000	160 000
Administration	130 000	140 000	150 000	160 000	80 000	120 000
Selling and distribution	140 000	160 000	180 000	200 000	80 000	120 000
Miscellaneous	30 000	40 000	50 000	60 000	20 000	20 000
	940 000	1 220 000	1 500 000	1 820 000	300 000	420 000

Example (Continued)

Some additional information is available:

1. Present sales of 50% capacity are estimated at €600,000 per annum.
2. Estimated costs of closing down are €90,000. In addition, maintenance of plant and machinery is expected to amount to €16,000 per annum.
3. Costs of reopening after being closed down would be approximately €40,000 for overhauling machines and getting ready and €28,000 for training of key personnel.
4. Economic indicators suggest that sales should take an upward swing to around 70% capacity at prices which would reflect a dramatic uplift, producing revenue of €2,000,000 in 12 months' time.

Two alternatives present themselves: to keep the chain division open during the recession or to shut down and wait for the economic upturn

It can be seen that there are only three relevant levels of operations: zero % (i.e. close down and reopen), 50% (present level), and 70% (post-recession level). Revenues and costs should therefore be set out for all three levels

	Level of operations in €		
	0%	50%	70%
Sales	–	€600 000	€2 000 000
Marginal costs	–	660 000	940 000
Contribution	–	€(60 000)	€1 060 000
Fixed costs	€300 000	420 000	420 000
Special costs	174 000	€474 000	
Gain/(Loss)		€(474 000)	€640 000

Calculations

Marginal Costs are calculated by:

- (a) Interpolation, i.e. for 50%, the mid-point between 40% and 60%; for 70%, the mid-point between 60% and 80%
- (b) Deducting the fixed components of factory overheads, administration, selling and distribution and miscellaneous from the total figures to determine the marginal cost component

Marginal costs of operations	@50%	@70%
Mid-point of flexible budget:		
@ 50% €1 220 000 – €940 000	€1 080 000	
@ 70% €1 500 000 – €1 220 000		€1 360 000
Deduct fixed costs	420 000	420 000
	<u>€660 000</u>	<u>€940 000</u>

The special costs of closing down and reopening are as follows.

$$€90\,000 + €16\,000 + €40\,000 + €28\,000 = €174\,000$$

Conclusions

Focusing on Contribution has benefits

Management can gain a useful picture of these costs which vary with volume changes, and those that do not

Focusing on Contribution poses risks

Management might ignore the fixed component of the factory overheads, administration, selling and distribution, and miscellaneous costs and treated all costs as variable costs



This would have produced totally different figures and would have led to wrong decisions being taken

It's hard to decide based on the previous information whether to close or not

If the division is kept open at a 50% level of operations, the division would incur only €6,000 greater loss, a cost which the management of Clydeside may well be prepared to incur to ensure harmonious industrial relations

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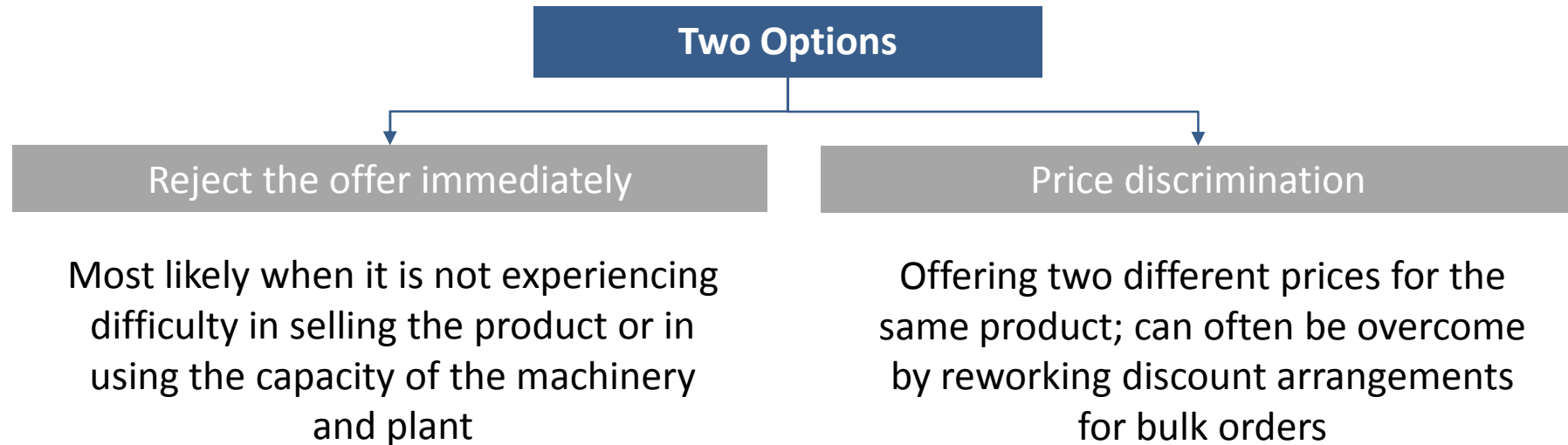
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It is usual for a company to be approached by a potential customer with an offer to buy a large quantity of product at an advantageous price



Companies need a framework for analyzing this type of offer

Example

The Highland Shortbread Company was experiencing a decline in sales due to the economic recession in the home market; consumers were buying only basic provisions for their families and were consequently giving the supermarkets' confectionery displays a wide berth. An increase in sugar prices had also caused a fairly steep increase in the retail price of shortbread.

A British national airline approached the company with an offer to buy a special order of shortbread to enable them to serve passengers with a complimentary cup of coffee and finger of shortbread on overseas flights terminating in Scotland. If the experiment met with customer approval the airline would seek to enter into a long-term contract with the Highland Shortbread Company. The special order was for 1,000 cartons of shortbread at €10 per carton.

Example (Continued)

The sales director believed the price of €10 per carton was lower than even the cost of manufacturing one carton of shortbread, far less the cost, taking into account the cost of selling and distribution. To confirm his belief he called for a copy of the previous year's income statement which revealed the following picture:

Sales	€150 000
Manufacturing cost of goods sold	120 000
Gross margin	<u>€ 30 000</u>
Selling and administrative expenses	20 000
Operating income	<u>€ 10 000</u>

This income statement was based on sales of 10,000 cartons of shortbread. Hence the cost to manufacture one carton was €12 ($€120,000/10,000$). The sales director therefore recommended an outright rejection: 'There is no way we can subsidize the airline on these figures; their offer of €10 does not even cover the cost of manufacturing the stuff. We would lose €2,000 on the deal. If they want our shortbread, they can jolly well pay €15 per carton like everyone else.'

Example (Continued)

The managing director was not happy with the sales director's conclusions. Even without studying the figures he would have been happy to accept the special order because of the prospect of concluding a long-term contract with the airline. (Indeed, as he privately admitted, he would have given the airline the shortbread at no charge just to get the long-term business!) However, he called for a detailed breakdown of the income statement. This is what he found:

Sales		€150 000
Variable expenses		
Manufacturing	€90 000	
Selling and administration	5 000	95 000
Contribution margin		<u>€55 000</u>
Fixed expenses		
Manufacturing	30 000	
Selling and administration	15 000	45 000
Operating income		<u>€10 000</u>

Example (Continued)

The variable cost of manufacturing one carton of shortbread is



Only €9
(€90,000/10,000)

Each carton would be sold for



€10

Each carton would provide a contribution to fixed costs of



€10

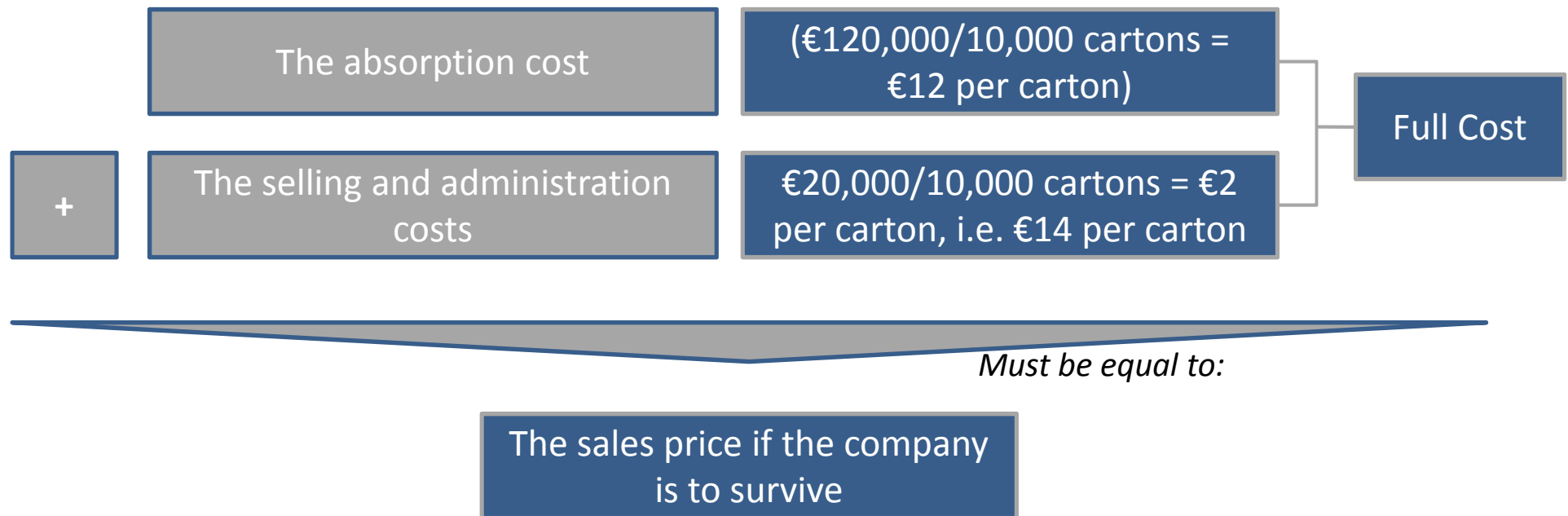
The special order would therefore provide a total contribution of €1,000

Since the fixed costs of €45,000 would not be affected by the airline's order, the company's overall profit would be increased by €1,000

The relevant costs in this example are only the variable costs

However, that the 'full cost' income statement used by the sales director is useful for long-term pricing purposes

The company must recover all of its costs, fixed and variable, eventually and must price accordingly



But management should not confuse costs required for long-term pricing strategies with those required to deliberate on special orders

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Processing products beyond split-off is a typical management problem in industries where the production process produces more than one product

Example

- In the petroleum industry where the refining of crude oil is undertaken to produce high quality petroleum oil, the process also produces a variety of valuable by-products which can be sold or processed further
- If these by-products become as valuable as the main product, they are called joint products

Example

Extending the wholesale butchery business example used earlier in this module, imagine that a food scientist, newly employed by the company, has suggested that the cheap cuts of shin beef and leg beef should be processed into both dog food and pie meat. He claims that both products would sell well, much better than selling the unprocessed cuts of meat. The processing costs, and separate sales prices are given in Figure below:

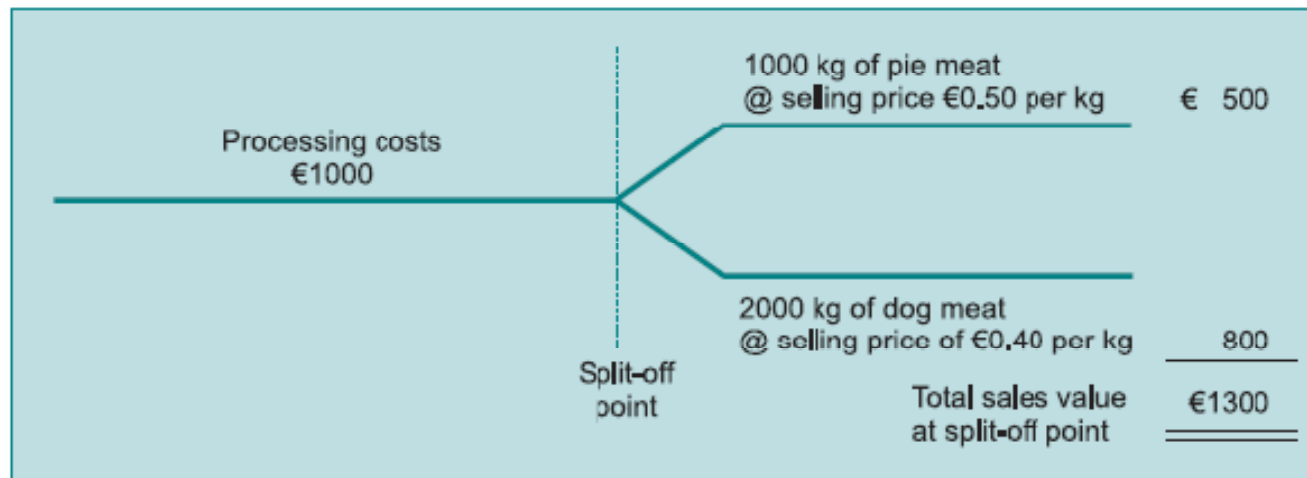


Figure 11.1 Processing costs and selling prices

We've said in module 10 that we'll allocate the processing costs using two methods: 1) Sales Value; and 2) Weight

Accountants deliberate long and hard about how to allocate the processing costs of €1,000. As we saw in Module 10 two methods are usually used. The weight method would allocate one-third of €1,000 to the pie meat and two-thirds to the dog meat. The sales value technique would allocate five-thirteenths of €1,000 to the pie meat and eight-thirteenths to the dog meat. This allocation procedure would be necessary for calculating inventory costs and therefore profit for the accounting period. Depending on the method chosen, different answers are produced

	Weight		Sales value	
	Pie meat	Dog meat	Pie meat	Dog meat
Sales	€500	€800	€500	€800
Allocation of joint cost	333	667	385	615
Profit	€167	€133	€115	€185
	€300		€300	

Plenty of arguments can be found to support either method and in the long term, it is argued, it does not matter which method is chosen because the profit is the same in both cases

Example (Continued)

Consider a further proposal by the food scientist that the company should make the pies instead of selling the pie meat to an outside pie-maker. His proposal envisages 1-kg plate pies, requiring €220 of pastry and further labor, which would sell for €0.60 each. If these new facts are fed into the income statements above, the following picture emerges.

	Plate Pies			
	By weight		By sales value	
Sales @ €0.60		€600		€600
Joint costs	€333		€385	
Additional costs	220	553	220	605
Profit/(Loss)		€47		€(5)

The decision would be taken to sell the pie meat in its bulk form to the outside pie-maker

This outcome is plainly ridiculous given that the same pie meat is to be processed in an identical manner. The only approach to the problem of further processing is the incremental approach where one compares the additional, or incremental revenue from sales with the incremental costs involved.

Incremental revenue 1000 pies @ €0.10	€100
Incremental costs	220
Net loss	<u>(€120)</u>

No technique for allocating joint product costs is applicable to management decisions of whether a product should be sold at the split-off point or processed further. Such decisions can be aided only by incremental or relevant cost analysis

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Multiple Choice Questions

11.1 Management has several reasons for requiring to know the total cost per unit of production. The principal reasons relate to decisions on:

- i. inventory valuation;
- ii. levels of planned production;
- iii. fixing selling prices;
- iv. salesmen's commission levels.

Which of the following is correct?

- A. (i) and (ii) only.
- B. (i) and (iii) only.
- C. (ii) and (iii) only.
- D. (ii), (iii) and (iv) only.

Multiple Choice Questions (Continued)

11.2 The composite elements of a predetermined overhead allocation rate mean that there is limited scope for accuracy because:

- i. applied overheads differ from actual overheads;
- ii. budgeted overheads differ from actual overheads;
- iii. budgeted output differs from applied output;
- iv. budgeted output differs from actual output.

Which of the following is correct?

- A. (i) only.
- B. (i) and (iii) only.
- C. (ii) and (iii) only.
- D. (ii) and (iv) only.

Multiple Choice Questions (Continued)

11.3 The principal difference between absorption costing and variable costing lies in the treatment of fixed manufacturing overhead which is included in product costs in absorption costing, but excluded from product costs in variable costing. True or false?

True

Multiple Choice Questions (Continued)

11.4 In a situation where production equals sales and planned production equals actual production, which of the following is correct?

- A. The contribution margin under variable costing equals the gross profit under absorption costing.
- B. The net profit under absorption costing equals the net profit under variable costing.
- C. The variable cost of sales under variable costing equals the full cost of sales under absorption costing.
- D. The full cost of sales under absorption costing equals the fixed production costs under variable costing.

Multiple Choice Questions (Continued)

The following information applies to Questions 11.5–11.9.

Energy Products Limited has prepared its cost budget for next year. Based on sales and production of 2,000 units projected costs are:

	€
Direct materials	46 000
Direct labour	34 000
Direct power	16 000
Variable manufacturing overhead	20 000
Fixed manufacturing overhead	30 000
Variable selling overhead	10 000
Fixed administrative overhead	18 000
	<u>174 000</u>
Each unit sells for €150.	

Multiple Choice Questions (Continued)

11.5 If variable costing is used, what is the inventory cost per unit?

- A. €48.
- B. €58.
- C. €73.
- D. €87.

	€000
Direct materials	46
Direct labour	34
Direct power	16
Variable manufacturing overhead	20
	116 / 2000 units = €58 per unit

Multiple Choice Questions (Continued)

11.6 If absorption costing is used, what is the inventory cost per unit?

- A. €58.
- B. €65.
- C. €73.
- D. €82.

	€000
Direct materials	46
Direct labour	34
Direct power	16
Variable manufacturing overhead	20
Fixed manufacturing overhead	30
	146 / 2000 units = €73 per unit

Multiple Choice Questions (Continued)

11.7 If absorption costing is used, what is the gross profit per unit?

A. €77.

B. €82.

C. €87.

D. €92.

$$\begin{aligned}\text{Gross profit} &= \text{Selling price} - \text{Product cost} \\ &= €150 - €73 \\ &= €77\end{aligned}$$

Multiple Choice Questions (Continued)

11.8 If variable costing is used, what is the total contribution margin?

A. €154,000.

B. €174,000.

C. €184,000.

D. €204,000.

$$\begin{aligned}\text{Total contribution margin} &= \text{Unit contribution margin} \times \text{Volume} \\ &= (\text{Selling price} - \text{Variable cost}) \times 2000 \text{ units} \\ &= (\text{€}150 - \text{€}63) \times 2000 \text{ units} \\ &= \text{€}87 \times 2000 \\ &= \text{€}174\,000\end{aligned}$$

Note: Variable cost per unit comprises:

Inventory cost per unit	=	€58
Variable selling overhead		
€10 000/2000 units	=	<u>5</u>
		<u>€63</u>

Multiple Choice Questions (Continued)

11.9 If only 1,800 units were sold out of 2,000 manufactured, which of the following is correct?

- A. Net profit under absorption costing exceeds net profit under variable costing by €2,000.
- B. Net profit under variable costing exceeds net profit under absorption costing by €2,000.
- C. Net profit under absorption costing exceeds net profit under variable costing by €3,000.
- D. Net profit under variable costing exceeds net profit under absorption costing by €3,000.

Multiple Choice Questions (Continued)

Absorption costing	€	€
Sales (1800 × €150)		270 000
Less: Full cost of sales (1800 × €73)		<u>131 400</u>
Gross profit		138 600
Other expenses:		
Variable selling overhead (1800 × €5)	9 000	
Fixed administrative overhead	<u>18 000</u>	<u>27 000</u>
		111 600
Variable costing	€	€
Sales (1800 × €150)		270 000
Less: Variable cost of sales (1800 × €63)		<u>113 400</u>
		156 600
Less: Fixed costs		
Fixed manufacturing overhead	30 000	
Fixed administrative overhead	<u>18 000</u>	<u>48 000</u>
		108 600
Difference in net profit = €3000		

Multiple Choice Questions (Continued)

11.9 If only 1,800 units were sold out of 2,000 manufactured, which of the following is correct?

- A. Net profit under absorption costing exceeds net profit under variable costing by €2,000.
- B. Net profit under variable costing exceeds net profit under absorption costing by €2,000.
- C. Net profit under absorption costing exceeds net profit under variable costing by €3,000.
- D. Net profit under variable costing exceeds net profit under absorption costing by €3,000.

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Case Study 11.1

Holmes and Grant Limited is a producer of specialist opto-electronic components for motor vehicle manufacturers. Following extensive technical discussions which have generated an agreed specification, the motor vehicle manufacturer has offered a contract for the supply of 2,500 components over the next six months at a price of €190 each.

The following data are available on the specification of the product.

(a) Labor specification

Each component will need six hours of skilled labor and five hours of semi-skilled labor. Skilled labor is available and is paid €6 per hour. However, an additional employee will have to be recruited at a rate of €5 per hour for the work which would otherwise be carried out by the skilled labor. The current rate of pay for semi-skilled labor is €4 per hour and it will be essential to appoint an additional semi-skilled employee to perform this work.

Case Study 11.1 (Continued)

(b) Materials specification

There are three elements to the materials specification:

1. Part No. 542A – two per component

It is estimated that this special part can be purchased for €30 each, but the supplier will require a minimum order for 5,500 parts. The company has no other use for this part.

2. Material K6 – 4 kg per component

Material K6 is continuously used by the company in other products. Current stockholdings amount to 4,000 kg, purchased at €2.35 per kg. The supplier has indicated that a price rise is imminent to €3.00 per kg.

3. Material J4 – 2 kg per component

Current stockholdings amount to 5,000 kg, acquired originally for €5.00 per kg. However, material J4 has not been in use for three years and it has been written down to a book value of €2.50 per kg. There is a proposal that, if J4 is subject to some extra processing at a cost of €2.00 per kg, then it could be used to substitute for another material L5 (in current use), which would otherwise have to be purchased for €4.50 per kg.

Case Study 11.1 (Continued)

(c) Overheads

Holmes and Grant Limited operates with an overhead allocation rate of 250 per cent on direct labor cost. Of 250 per cent total, 100 per cent is regarded as being a variable overhead and the remainder as fixed overhead. The particular requirements of this contract necessitate some extra fixed costs of €5,000.

Required

From the information available, assess whether this contract is acceptable. Pay particular attention to the relevant costs and support your view with a quantitative presentation to management.

Case Study 11.1 (Solution)

The relevant costs for the production of 2500 components are as follows:

		€	Note
Labour			
Skilled	(2500 × 6 hours × €5)	75 000	(1)
Semi-skilled	(2500 × 5 hours × €4)	50 000	(2)
Materials			
542A	(5500 × €30)	165 000	(3)
K6	(2500 × 4 kg × €3.00)	30 000	(4)
J4	(2500 × 2 kg × €2.50)	12 500	(5)
Overhead			
Variable only: 100% of labour cost, 100% of €125 000		125 000	
Specific extra fixed costs		5 000	(6)
		462 500	
Contract price (2500 × €190)		€475 000	
Contribution		€12 500	

Since the incremental income from the contract exceeds the incremental relevant costs, the advice to management is to accept the contract.

Case Study 11.1 (Solution)

Notes

1. Replacement cost is critical, as it will cost the company €5 per hour to replace the skilled labor, and this is the relevant cost for the contract.
2. As in Note 1 above, it is replacement cost which is relevant. In this case, it is the same as the current rate for semi-skilled labor.
3. Although only 5000 units of Part No. 542A are required to complete this contract, the fact that the supplier has insisted upon a minimum order of 5500 units means that the extra 500 will be surplus to requirements since the company does not use it elsewhere. Accordingly, the relevant cost is the full price of 5500 units.
4. The fact that the company already has a stockholding of material K6 is convenient, but irrelevant. This material is in constant use for other products. The relevant cost is the replacement cost of €3.00 per kg.

Case Study 11.1 (Solution)

Notes (Continued)

5. The company's original acquisition price of €5.00 per kg is irrelevant to this decision, as it is a sunk cost. Similarly, the lower written-down value of €2.50 per kg is also irrelevant, representing purely an accounting adjustment. If the material is not used on the contract, it will be utilized as a substitute for material L5. Using J4 as a replacement for L5 produces a saving of €2.50 per kg – the difference between the L5 purchase price of €4.50 and the further processing costs of €2.00. Therefore, the relevant cost of J4 consists of the opportunity cost of €2.50 per kg.
6. It is a rare situation when fixed costs are part of relevant costs. This is the case because there is a specific increase in fixed costs related to the acceptance of this contract.

Case Study 11.2

Hasting Bros Ltd is a small manufacturing company which produces a range of three special cashmere sweaters under the brand-names of Alpine, Border and Island. For April, the Management Accountant has prepared the following forecast of trading results:

	Alpine €000	Border €000	Island €000	Total €000
Sales	1 000	960	320	2 280
Variable costs				
Direct materials	350	300	100	750
Direct labour	50	80	30	160
Works overhead	200	180	110	490
Fixed overhead (apportioned)	300	270	100	670
	900	830	340	2070
Net Profit/(Loss)	100	130	(20)	210

Case Study 11.2 (Continued)

The following information is also available:

1. Although there are other materials which are required for the manufacture of the cashmere sweaters, the major material is cashmere, purchased directly from China. Unfortunately, due to delays in shipment from China, April production will be limited by the availability of cashmere supplies. The Purchasing Manager believes that he will be able to increase the current stockholding of 61,000 kg by a delivery of 31,000 kg in mid-April. There are no substitute suppliers in the short term.
2. The Production Manager advises that the different products absorb different quantities of cashmere, as follows:

	Cashmere per unit
Alpine	8 kg
Border	4 kg
Island	1 kg

3. The Sales Manager confirms that the current unit prices of the products are:

	€
Alpine	100
Border	120
Island	80

Case Study 11.2 (Continued)

He is convinced that sales of the Border product could be substantially increased beyond the current forecast, although he is unable to quantify the effect. Extra advertising of €80,000 would be required to achieve this increase, together with a 10 per cent reduction in the price of the Alpine product. The current forecast of trading results does not include this view.

4. The Managing Director has reviewed the forecast for April and believes that results can be improved immediately through stopping the manufacture of the Island product. Neither the Sales Manager nor the Production Manager agrees with his view, but they are not certain why they disagree.
5. Fixed overhead is apportioned over the product lines on the basis of an allocation by space occupied on the factory floor. The Managing Director is confused by the different proposals being put forward and he calls on your services as the company's external financial consultant to assist him in drawing up a sensible plan of action.

Case Study 11.2 (Continued)

Required

1. Assuming that the full sales forecast for April can be achieved, assess the impact of the Managing Director's proposal to drop the entire Island range and advise on its desirability.
2. In view of the shortage of cashmere, assist the company by preparing:
 - (a) an optimal production plan for April;
 - (b) a revised forecast of trading results for April.
3. In light of 2 above, assess the viability and effects of the Sales Manager's plans to increase sales of the Border range through extra advertising expenditure.
4. What other points of commercial interest would you wish to draw to the attention of the Managing Director?

Case Study 11.2 (Solution)

1

- The Managing Director believes that the act of dropping the Island range of products will increase profits by €20,000 – since it seems that the loss forecast would be avoided.
- Unfortunately, it is necessary to appraise him (as tactfully as possible) that his thinking is being unduly influenced by the arbitrary allocation of €100,000 of fixed overheads to the Island range. If the Island range was stopped, then €100,000 of fixed overhead would not be ‘saved’, but merely reallocated to the other two lines.
- The correct approach is to assess the contribution to fixed overheads arising from the Island product, as follows:

	€000
Sales	320
Variable costs	
Direct materials	100
Direct labour	30
Works overhead	110
	240
Contribution	80

- In fact, therefore, if this proposal had been adopted, the forecast trading profit of €210,000 would have fallen to €130 000 due to this lost contribution. Hence, this proposal should be soundly rejected.

Case Study 11.2 (Solution)

2

- Since the shortage of cashmere constitutes a limiting factor on April's output, it is necessary to deploy the available cashmere over the three products – in a manner which will optimize overall profit. In the first instance, it is necessary to identify the ranking of the products, in terms of the contribution they generate:

	Alpine	Border	Island
	€000	€000	€000
Sales	1 000	960	320
Variable costs			
Direct materials	350	300	100
Direct labour	50	80	30
Works overhead	200	180	110
	600	560	240
Total contribution	400	400	80
Unit selling price	€100	€120	€80
Therefore, no. of units	10 000	8 000	4 000
Contribution margin (per unit)	€40	€50	€20
Cashmere (per unit)	8kg	4kg	1kg
Contribution per cashmere kg	€5.00	€12.50	€20.00
Ranking	3	2	1

- The ranking exercise indicates that the Island product (contrary to the Managing Director's belief) is the 'best' product in the current situation of limited supplies of cashmere.

Case Study 11.2 (Solution)

2

- As a result, the optimal production plan is:

Product	Forecast units	Cashmere required (kgs)	Unit contribution margin €	Total contribution €000
Island	4 000	4 000	20	80
Border	8 000	32 000	50	400
Alpine	7 000	(1) 56 000	40	280
		<u>92 000</u>		<u>760</u>
			Fixed costs	<u>670</u>
			Net profit	<u>90</u>

Note (1) As the third-ranked product, Alpine received only the residual balance of the available cashmere – 56 000 kgs. This means that only 7 000 units can be manufactured, leaving an unsatisfied demand for 3 000 units in April.

Case Study 11.2 (Solution)

3

- There are various aspects of the Sales Manager's plans to consider:
 - (a) There is no point in cutting the price of the Alpine range since there is already unsatisfied demand of 3,000 units in April. A price reduction of 10% will result in lost contribution of €70,000 ($€10 \times 7,000$) and reduce profits to €20,000.
 - (b) Based on the unit contribution margin for Border of €50, an extra €80,000 expenditure on advertising would require the sale of a minimum of an extra 1600 units ($€80,000/€50$). However, a further 1,600 units of Border would need a further 6,400 kg of cashmere – and hence result in less cashmere being available to manufacture the Alpine. At 8 kg per unit of Alpine, this would produce a reduction of 800 units in Alpine sales, and a lost contribution of €32,000 ($800 \times €40$)

Case Study 11.2 (Solution)

4

Other points of commercial interest include:

- (a) What can be done to increase sales of the Island range since it is the best performing product?
- (b) Is it possible to increase the selling prices of all the products as a means to restrict demand to attainable levels?
- (c) Is there any impact on sales caused by the restricted availability of any one product?
- (d) Would any customers be prepared to wait for deliveries – until cashmere is in greater supply?